Neural Networks 171 (2024) 429–439

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

Dual-domain strip attention for image restoration

Yuning Cui *, Alois Knoll

School of Computation, Information and Technology, Technical University of Munich, Munich, 85748, Germany

ARTICLE INFO

Keywords:
Image restoration
Dual-domain learning
Efficient network

ABSTRACT

Image restoration aims to reconstruct a latent high-quality image from a degraded observation. Recently, the usage of Transformer has significantly advanced the state-of-the-art performance of various image restoration tasks due to its powerful ability to model long-range dependencies. However, the quadratic complexity of self-attention hinders practical applications. Moreover, sufficiently leveraging the huge spectral disparity between clean and degraded image pairs can also be conducive to image restoration. In this paper, we develop a dual-domain strip attention mechanism for image restoration by enhancing representation learning, which consists of spatial and frequency strip attention units. Specifically, the spatial strip attention unit harvests the contextual information for each pixel from its adjacent locations in the same row or column under the guidance of the learned weights via a simple convolutional branch. In addition, the frequency strip attention unit refines features in the spectral domain via frequency separation and modulation, which is implemented by simple pooling techniques. Furthermore, we apply different strip sizes for enhancing multi-scale learning, which is beneficial for handling degradations of various sizes. By employing the dual-domain attention units in different directions, each pixel can implicitly perceive information from an expanded region. Taken together, the proposed dual-domain strip attention network (DSANet) achieves state-of-the-art performance on 12 different datasets for four image restoration tasks, including image dehazing, image desnowing, image denoising, and image defocus deblurring. The code and models are available at https://github.com/c-yn/DSANet.

1. Introduction

Image restoration aims to reconstruct a latent sharp image from its degraded counterpart, essential for surveillance, remote sensing, and medical imaging. To solve this ill-posed problem, conventional methods resort to various assumptions and hand-crafted features. However, these approaches are incapable of generating faithful results when the requirements are not satisfied, and hence, they are impractical for more complicated real-world scenarios (Zhang, Ren, et al., 2022).

Recently, convolutional neural networks (CNNs) have achieved more advanced performance for image restoration over traditional algorithms by learning generalizable priors from collected large-scale datasets (Cho, Ji, Hong, Jung, & Ko, 2021; Lee, Son, Rim, Cho, & Lee, 2021; Tian, Fei, et al., 2020; Tian, Xu, & Zuo, 2020). Many modules have been developed or borrowed from other domains to further boost the performance, such as encoder–decoder architecture (Cui, Tao, Jing, & Knoll, 2023), dilated convolution (Zou et al., 2021), and attention (Qin, Wang, Bai, Xie, & Jia, 2020; Tian, Xu, Li, et al., 2020; Zhang, Wang, Wang, & Jiang, 2021). Nevertheless, convolution operators have two weaknesses that are not matched with image restoration: *i*) The static filters of convolution operators are not adaptive to input features, which is inapplicable to spatially varying blurs; *ii*) The limited receptive fields cannot model long-range pixel interactions for large-scale degradations.

More recently, inspired by their successful applications in high-level vision tasks, Transformer models have been introduced into low-level vision tasks and significantly advanced state-of-the-art performance. The core element, self-attention, can model long-range dependencies effectively (Cui & Knoll, 2023). Nonetheless, its quadratic complexity with respect to the related dimension makes it infeasible for image restoration, which always involves high-resolution images. To alleviate this problem, a few remedies have been proposed to improve the efficiency of image restoration. For example, Liang, Cao, Sun, Zhang, Van Gool, and Timofte (2021) computed self-attention within small windows. Zamir et al. (2022) applied self-attention to the channel dimension instead of the spatial dimension. Tsai, Peng, Lin, Tsai, and Lin (2022) developed strip-based self-attention for image deblurring. Despite the reduced complexity, these methods do not break the nature of self-attention, *i.e.,* they still have quadratic complexity to the size of windows, channels, or strips.

In this paper, we develop an efficient and effective network for image restoration by proposing a dual-domain strip attention mechanism.

* Corresponding author.
E-mail address: yuning.cui@in.tum.de (Y. Cui).

https://doi.org/10.1016/j.neunet.2023.12.003
Received 18 September 2023; Received in revised form 28 October 2023; Accepted 1 December 2023
Available online 4 December 2023

More concretely, the spatial strip attention unit performs information aggregation for each pixel from its adjacent locations in the same row or column. This process is guided by attention weights learned via a simple convolution branch. Furthermore, to reduce frequency gaps between sharp and degraded image pairs, the frequency strip attention unit refines spectra by decoupling features into different frequency components and then modulating them with lightweight attention weights. To enhance multi-scale representation learning and better deal with degradations of various sizes, we apply different strip sizes to group-wise features. With joint horizontal and vertical information aggregation, each pixel can perceive signals from a large region centered on itself.

The proposed strip attention mechanism enjoys several key advantages: **(a)** It modulates features in two domains with low computation budget; **(b)** By disintegrating attention into two directions, it improves efficiency while remaining implicit large-scale receptive fields; **(c)** It is capable of capturing multi-scale contextual information; **(d)** It is content-aware to adapt to different input features. Our spatial attention is significantly distinguished from strip-based Transformer approaches (Dong et al., 2022; Li, Fan, et al., 2023; Tsai et al., 2022). These algorithms execute self-attention within strip-shaped regions, which entail quadratic complexity. Differently, our method generates attention weights from a simple bypass network and conducts integration in a cheap manner. In addition, we exploit dual-domain and multi-scale representation learning in our design. The main contributions of the paper are summarized as follows:

- We propose a spatial strip attention unit that integrates contextual information efficiently.
- We develop a frequency strip attention unit that refines spectra via frequency separation and modulation using extremely simple strip pooling techniques.
- We introduce the dual-domain strip attention mechanism that executes dual-domain two-directional attention, efficiently enhancing representation learning at multiple scales for image restoration.
- The proposed network, dubbed DSANet, achieves state-of-the-art performance on 12 benchmark datasets for four image restoration tasks.

This paper is an extension of the conference work (Cui, Tao, Jing, & Knoll, 2023). The main improvements over the preliminary version are as follows: **(a)** We propose the spectral strip attention unit to refine spectra at multiple scales by decoupling features into different frequency components via strip-shaped pooling techniques and modulating them with lightweight learnable parameters. This modification leads to consistent improvements on three image restoration tasks with low computation overhead. Specifically, our method improves the performance by 0.38 dB PSNR on SOTS-Outdoor (Li et al., 2018) with only extra 0.05M parameters and 0.46 GFLOPs. **(b)** Our method is extended to the image denoising task. In addition, our model is further evaluated on seven datasets, including four real-wold dehazing datasets, a nighttime dehazing dataset, and two desnowing datasets. **(c)** More ablation studies are conducted to demonstrate the effectiveness of the newly designed spectral attention.

2. Related work

2.1. Image restoration

Images captured under bad conditions or with low-end devices will degrade the visibility and impact the applications in downstream vision tasks. As a long-standing and challenging task, image restoration aims to reconstruct a latent sharp image from a distorted observation with undesired degradations, such as blur, snowflakes, and haze. To resolve this ill-posed problem, many conventional algorithms have been developed based on hand-crafted features and assumptions to obtain plausible results. Nevertheless, these approaches are impractical for more complex real-world scenarios (Zhang, Ren, et al., 2022).

Recently, with the rapid development of deep learning, manifold pioneering CNN-based methods have been proposed by learning generalizable priors from collected large-scale datasets and provided promising performance on image restoration tasks. For instance, (Cai, Xu, Jia, Qing, & Tao, 2016) proposed a trainable end-to-end network that takes a hazy image as input and outputs its medium transmission map, which is subsequently used to produce a haze-free image via the atmospheric scattering model (Narasimhan & Nayar, 2000). Nah, Hyun Kim, and Mu Lee (2017) developed a multi-scale CNN to recover sharp images in an end-to-end manner where motion blurs are caused by various sources. Fu, Huang, Ding, Liao, and Paisley (2017) leveraged CNN to directly learn the nonlinear mapping function between clean and rainy detail layers. Thereafter, many frameworks have been developed to further improve the performance of image restoration by deploying advanced modules. Among these designs, the U-shaped architecture is widely adopted to efficiently learn hierarchical representations (Cho et al., 2021). Dilated convolutions provide large and multi-scale receptive fields (Zou et al., 2021). Skip connections have been proven effective for learning residual signals (Lee et al., 2021). Moreover, various attention modules, *e.g.,* pixel, spatial, and channel attention, have been introduced to attend to informative information (Qin et al., 2020; Zamir et al., 2021).

More recently, Transformer-based models have been introduced into low-level vision tasks to pursue long-range pixel interactions and advanced state-of-the-art performance. Chen, Wang, et al. (2021) introduced a Transformer-based method for image processing to exploit the potential of Transformer by pre-training on large-scale synthesized ImageNet (Deng et al., 2009). Guo et al. (2022) firstly introduced the power of Transformer into image dehazing. To reduce the high complexity of self-attention, a few measures have been taken. Liang et al. (2021) and Wang, Cun, et al. (2022) restricted operation regions of self-attention to improve efficiency. Zamir et al. (2022) switched self-attention from the spatial dimension to the channel dimension. However, these remedies still have the quadratic complexity to the relevant dimension of input. In this paper, instead of designing advanced modifications for self-attention, we resort to CNN-based methods combined with the adaptive property of self-attention for efficient information aggregation.

2.2. Attention mechanisms

Inspired by their successful applications in high-level vision tasks, attention mechanisms have been introduced to image restoration to selectively attend to important information. Qin et al. (2020) leveraged channel attention and pixel attention to flexibly deal with different types of information for image dehazing. Liu, Ma, Shi, and Chen (2019) utilized channel-wise attention to adjust the contributions of different streams for feature fusion. Zamir et al. (2021) proposed the supervised attention module for feature filtering. Chen, Chu, Zhang, and Sun (2022) devised a simplified channel attention unit and a simple gate module for image restoration. These methods do not aggregate contextual information when imposing learned attention weights on features.

Another line on this topic is to design efficient self-attention for image restoration. Since our attention mechanism falls into the strip-based category, we mainly introduce strip-based self-attention modules. Tsai et al. (2022) proposed intra-/inter-strip attentions for dynamic scene image deblurring. Chen, Zhang, et al. (2022) developed the rectangle-window self-attention mechanism for image restoration. Li, Fan, et al. (2023) introduced anchored strip self-attention for long-range dependency modeling. Different from these strip-based self-attention modules that have quadratic complexity to strip sizes, our spatial strip attention unit computes attention weights and performs aggregation via cheap convolution. Additionally, we propose a frequency strip attention unit to refine features in the spectral domain.

2.3. Frequency-based networks

In the context of image restoration, high-frequency signals represent image textures and fine details, while low frequency is mainly related to flat and smooth areas. It is intuitive to recover a clean image in the spectral domain by treating different frequency components separately. In addition, the fast Fourier transform (FFT) is capable of efficiently modeling global features according to the convolution theorem. Taking these advantages into consideration, a few frequency-based networks have been proposed for image restoration based on FFT. Concretely, Guo, Yong, Zhang, Ma, and Zhang (2023) presented a window-based frequency channel attention to model long-range dependencies and resolve the frequency resolution mismatch problem. Huang et al. (2022) proposed to restore the phase and amplitude representations in different sub-networks. Li, Guo, et al. (2023) handled luminance and noises in the Fourier domain for low-light image enhancement. Mao, Liu, Shen, Li, and Wang (2021) enabled both low- and high-frequency learning by using convolutions after converting spatial features to the spectral domain.

In addition to FFT, there are some works using wavelet transform for image restoration. Specifically, Huang et al. (2023) presented a wavelet-based diffusion model to learn the distribution of sharp images in the wavelet domain. Zou et al. (2021) proposed to process wavelet components using different convolution layers. Chen, Fang, et al. (2021) applied the dual-tree wavelet transform and designed a wavelet loss to better present the complex snow shape. Furthermore, a few methods used other techniques for frequency processing, such as filters (Fu et al., 2017; Liu, Yeh, Chung, & Chang, 2020), OctConv (Chen et al., 2019; Zhang, Li, et al., 2022) and discrete cosine transform (Zhu, Zhang, Wang, Feng, & Duan, 2023). In this paper, we apply strip pooling tools to separate features into different frequency components in two directions and then modulate them via lightweight learned attention weights without using any convolutional layers.

3. Methodology

In this section, we first introduce the pipeline of the proposed network. Then, we delineate the architecture of the spatial strip attention unit (SSA) and frequency strip attention unit (FSA), respectively. Next, we describe the details of the dual-domain strip attention mechanism (DSAM). Finally, we introduce the used loss functions.

3.1. Overall pipeline

The overall pipeline of the proposed CNN-based DSANet is illustrated in Fig. 1(a). Our network adopts the popular encoder–decoder architecture to efficiently learn hierarchical representations and consists of three scales. There are three ResGroups in both the encoder and decoder sub-networks. ResGroup comprises N residual blocks and a customized one that accommodates the proposed DSAM between two 3×3 convolutions.

Given a degraded image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$, a single 3×3 convolution layer is used to extract shallow features of size $C \times H \times W$, where C is the number of channels and $H \times W$ denotes spatial locations. Then, the resulting features are fed into three ResGroups to obtain the in-depth features. During this process, the number of channels is expanded, while the spatial resolution is progressively reduced to $\frac{H}{4} \times \frac{W}{4}$. The downsampling layer is implemented by a strided convolution. Next, the deepest features with the lowest resolution pass through the decoder network to yield high-resolution representations. A transposed convolution is used for the upsampling layer. Moreover, the decoder features are concatenated with the encoder features as prior arts (Cui, Tao, Jing, & Knoll, 2023; Zamir et al., 2022) to assist in recovery, which is followed by a 1×1 convolution to reduce channels by half. Finally, a 3×3 convolution is applied to high-resolution features to produce the learned residual image $\mathbf{R} \in \mathbb{R}^{3\times H\times W}$, to which the degraded image is added to generate the restored image via $\hat{\mathbf{I}} = \mathbf{I} + \mathbf{R}$.

3.2. Spatial Strip Attention Unit (SSA)

Since the canonical self-attention module inspires our spatial strip attention unit, we first provide complexity analyses of self-attention before describing the formulation of our unit. Given an input tensor $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, self-attention can be formally expressed as:

$$\begin{aligned}&\mathrm{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{Softmax}(\mathbf{Q}\mathbf{K}^{\top})\mathbf{V},\\ &\text{where}\quad \mathbf{Q} = \mathbf{X}\mathbf{W}^{Q}, \mathbf{K} = \mathbf{X}\mathbf{W}^{K}, \mathbf{V} = \mathbf{X}\mathbf{W}^{V}.\end{aligned} \tag{1}$$

Here, we omit the normalization term for simplicity. We can observe from Eq. (1) that the complexity of self-attention comes from three aspects: *i*) the production of query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) tensors with the complexity of $3HWC^2$; *ii*) the generation of the attention map based on key-query dot-product with the complexity of $(HW)^2C$; and *iii*) the weighted summation process with the complexity of $(HW)^2C$. We can see that the complexity is quadratic to the spatial size in the last two terms.

Our SSA is designed for efficient information aggregation from the perspective of reducing the complexity of the three steps mentioned above. SSA consists of vertical and horizontal strip attention operators. Since two-directional operators share a similar pipeline, we only introduce the details of the horizontal one. Concretely, as illustrated in Fig. 2(a), given any input features $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, we remove the procedure of generating $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$, and instead directly yield attention weights via an extremely lightweight branch, which consists of global average pooling (GAP), a 1×1 convolution, and Sigmoid function. The attention weights generation process can be formally expressed as:

$$\mathbf{A} = \sigma(W_{1\times 1}(\mathrm{GAP}(X))) \in \mathbb{R}^{K}, \tag{2}$$

where $W_{1\times 1}$ denotes a 1×1 convolution; σ represents the Sigmoid function; and $K (K \ll W)$ specifies the length of a horizontal strip for integration. We share $\mathbf{A}$ across spatial and channel dimensions for further efficiency.

Then, the refined features can be obtained via the convolution-style integration method, which is formally expressed as:

$$\hat{\mathbf{X}}_{c,h,w} = \sum_{k=0}^{K-1} \mathbf{A}_k \mathbf{X}_{c,h,w-\lfloor \frac{K}{2} \rfloor + k} \tag{3}$$

Our aggregation method has a complexity of $HWCK$, which is much lower than that of self-attention with $(HW)^2C$.

To summarize, the horizontal spatial strip attention operator can be formally expressed as:

$$\hat{\mathbf{X}} = S_K^{\mathrm{H}}(\mathbf{X}) \tag{4}$$

It has been illustrated in previous works (Wang, Cun, et al., 2022; Zamir et al., 2022) that enlarging the receptive field of the network is beneficial to image restoration. Motivated by this fact, the output of our SSA is obtained by sequentially using horizontal and vertical strip attention operators as:

$$\mathbf{X}_K^{\mathrm{SSA}} = \mathrm{SSA}_K(\mathbf{X}) = S_K^{\mathrm{V}}(S_K^{\mathrm{H}}(\mathbf{X})), \tag{5}$$

where $S_K^{\mathrm{V}}(\cdot)$ denotes the vertical strip attention operator. By doing this, SSA implicitly enlarges the receptive field of the network. As illustrated in Fig. 3, the horizontal and vertical attention operators perform information integration in two directions, respectively. For convenience, we only pick a few representative pixels for illustration. The horizontal one gives $\mathrm{B} = w_{AB}\mathrm{A} + w_{BB}\mathrm{B} + w_{CB}\mathrm{C}$, where w_{ij} denotes the attention weight from i to j. By using two-directional operators successively, the value of D in Fig. 3(b) is computed by:

$$\begin{aligned}\mathrm{D} &= w_{BD}\mathrm{B} + w_{DD}\mathrm{D}\\ &= w_{BD}(w_{AB}\mathrm{A} + w_{BB}\mathrm{B} + w_{CB}\mathrm{C}) + w_{DD}\mathrm{D}.\end{aligned} \tag{6}$$

Consequently, the pixel in the center implicitly perceives contexts from the whole region determined by $K \times K$.

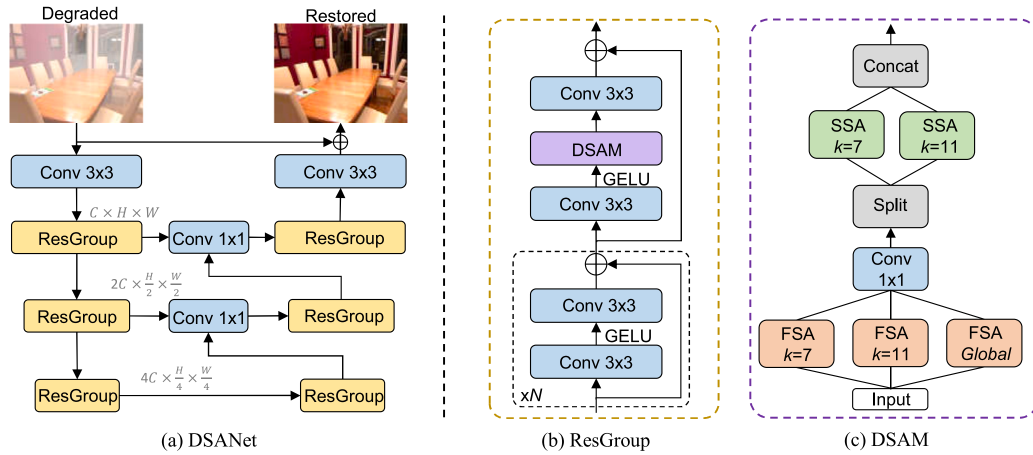

Fig. 1. The architecture of the proposed DSANet. K denotes the strip size.

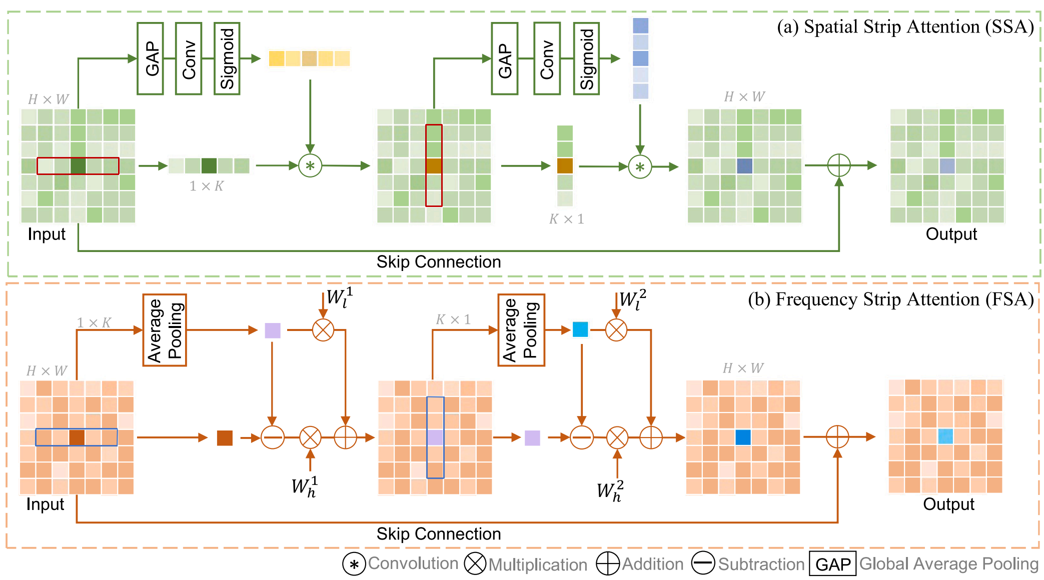

Fig. 2. The schematic diagrams of SSA (top) and FSA (bottom). We only showcase a single feature channel for brevity.

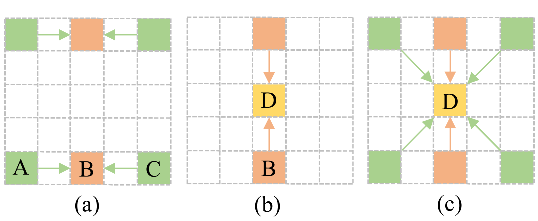

Fig. 3. Different integration paradigms. (a) Horizontal strip attention. (b) Vertical strip attention. (c) Spatial strip attention unit (SSA).

3.3. Frequency Strip Attention Unit (FSA)

In addition to SSA, we propose FSA to sufficiently utilize significant frequency disparities between degraded and clean image pairs via strip-based frequency separation and modulation. The schematic diagram is presented in Fig. 2(b). Similarly, we take the horizontal operator as an example for illustration. Specifically, given an input tensor $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, we firstly leverage strip-shaped average pooling as mean filters to obtain the low-frequency component, *i.e.,* lowest frequency, for each strip region $\mathbf{S}_{c,h,w} \in \mathbb{R}^{1\times K}$ centered at $\mathbf{X}_{c,h,w}$, where K is the length of strips. Then, the complementary high-frequency component can be obtained by removing the resulting low frequency from input features. Next, we impose channel-wise learnable attention parameters on different frequency components to enable the network to selectively attend to informative frequencies. The parameters are extremely lightweight and are directly updated by back-propagation. Note that in this process, we do not use any convolution layers as previous methods (Guo et al., 2023; Zou et al., 2021). Finally, the output of the horizontal operator is obtained by adding the refined frequencies together. The above process can be formally expressed as:

$$\mathbf{X}^{l}_{c,h,w} = \mathrm{AP}(\mathbf{S}_{c,h,w}) \tag{7}$$

$$\mathbf{X}^{h}_{c,h,w} = \mathbf{X}_{c,h,w} - \mathbf{X}^{l}_{c,h,w} \tag{8}$$

$$\hat{\mathbf{X}} = W_l^1 \mathbf{X}^l + W_h^1 \mathbf{X}^h \tag{9}$$

where W_l^1 and $W_h^1 \in \mathbb{R}^C$ are channel-wise attention parameters for modulating low- and high-frequency components, respectively. The above horizontal attention operator can be summarized as $\hat{\mathbf{X}} = \mathcal{F}^{\mathrm{H}}_{K}(\mathbf{X})$. Similar to SSA, our FSA is given by sequentially deploying horizontal and vertical operators:

$$\mathbf{X}^{\mathrm{FSA}}_{K} = \mathrm{FSA}_K(\mathbf{X}) = \mathcal{F}^{\mathrm{V}}_{K}(\mathcal{F}^{\mathrm{H}}_{K}(\mathbf{X})) \tag{10}$$

where $\mathcal{F}^{\mathrm{V}}_{K}(\cdot)$ represents the vertical operator with attention parameters of W_l^2 and W_h^2 as shown in Fig. 2(b).

3.4. Dual-Domain Strip Attention Mechanism (DSAM)

DSAM is designed for organizing SSA and FSA and improving multi-scale representation learning. As illustrated in Fig. 1(c), the input features first pass through three FSA with different strip sizes. Next, a 1×1 convolution is used to refine the added features of three frequency branches. The resulting features are split into two even parts among the channel dimension. Subsequently, two parts are fed into different SSA with $K = 7$ and $K = 11$ for spatial modulation. The final output of DSAM is obtained by concatenating resultant features. Formally, given any input $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, the process of DSAM can be expressed as:

$$\mathbf{X}' = W_{1\times 1}(\mathrm{FSA}_{K=7}(\mathbf{X}) + \mathrm{FSA}_{K=11}(\mathbf{X}) + \mathrm{FSA}_{global}(\mathbf{X})) \tag{11}$$

$$\mathbf{X}_1,\ \mathbf{X}_2 = \mathrm{Split}(\mathbf{X}') \tag{12}$$

$$\hat{\mathbf{X}} = [\mathrm{SSA}_{K=7}(\mathbf{X}_1), \mathrm{SSA}_{K=11}(\mathbf{X}_2)] \tag{13}$$

where $\mathbf{X}'$ and $\hat{\mathbf{X}}$ denote the intermediate result and final output of DSAM, respectively; *global* means $K = W/H$ in horizontal/vertical operators; Split and $[\cdot,\cdot]$ represent splitting and concatenating features among channels, respectively.

3.5. Loss function

To facilitate learning of the proposed dual-domain mechanism, we adopt the dual-domain L_1 loss (Cho et al., 2021) as:

$$\mathcal{L}_s = \frac{1}{E}\|\hat{\mathbf{I}} - \mathbf{G}\|_1 \tag{14}$$

$$\mathcal{L}_f = \frac{1}{E}\|\mathcal{F}(\hat{\mathbf{I}}) - \mathcal{F}(\mathbf{G})\|_1 \tag{15}$$

where $\hat{\mathbf{I}}$ and $\mathbf{G}$ denote the restored image and ground truth, respectively; E denotes the total elements for normalization; $\mathcal{F}$ represents the fast Fourier transform. The final loss function is given by combining the above two terms as $\mathcal{L}_{total} = \mathcal{L}_s + \lambda\mathcal{L}_f$, where λ is empirically set to 0.1.

4. Experiments

In this section, we introduce the universal implementation details and then detail the used datasets and specific training setups. Finally, we compare our method with state-of-the-art algorithms on 12 different benchmark datasets for four representative image restoration tasks, including image dehazing, image desnowing, image defocus deblurring, and image denoising. In the tables, the best and second-best results are in **bold** and underlined, respectively.

4.1. Implementation details and evaluation metrics

Unless stated otherwise, the following settings are adopted for training DSANet. Specifically, Adam is used as the optimizer with the initial learning rate as $2e^{-4}$, which is gradually reduced to $1e^{-6}$ with cosine annealing. We only apply horizontal flips with a probability of 0.5 for data augmentation. Models are trained on 256 × 256 patches with a batch size of 8. According to the task complexity, we scale the network by setting $N = 3$ (Fig. 1(b)) for dehazing/desnowing, $N = 15$ for defocus deblurring, and $N = 17$ for denoising. FLOPs are measured on a patch of size 3 × 256 × 256.

4.2. Datasets

4.2.1. Image dehazing

We evaluate the proposed method on daytime and nighttime datasets. The synthetic dataset (RESIDE (Li et al., 2018)) and real-world datasets (Dense-Haze (Ancuti, Ancuti, Sbert, & Timofte, 2019), O-HAZE (Ancuti, Ancuti, Timofte, & De Vleeschouwer, 2018), NH-HAZE (Ancuti, Ancuti, & Timofte, 2020) and NH-HAZE2 (Ancuti, Ancuti, Vasluianu, & Timofte, 2021)) are used for daytime scenarios. The RESIDE dataset consists of two subsets for training, *i.e.,* indoor training set (ITS) and outdoor training set (OTS). ITS contains 13,990 hazy images produced from 1,399 clean images. OTS comprises 313,950 hazy images generated from 8,970 hazy images. We train the proposed model on ITS and OTS separately and then apply the trained models to the corresponding test sets of RESIDE, *i.e.,* SOTS-Indoor and SOTS-Outdoor, both including 500 hazy images. The model is trained in indoor and outdoor scenes for 1,000 and 30 epochs, respectively. The training configurations for real-world datasets follow that of Guo et al. (2022) except the patch size, which is set as 500 × 700 in our experiments. The NHR (Zhang, Cao, Zha, & Tao, 2020) dataset is used for nighttime evaluation, which has 16,146 and 1,794 image pairs for training and testing, respectively. The model is trained for 300 epochs.

4.2.2. Image defocus deblurring

Consistent with prior arts (Lee et al., 2021; Zamir et al., 2022), we evaluate DSANet on the widely used DPDD (Abuolaim & Brown, 2020) dataset. This dataset includes images of 500 indoor/outdoor scenes, which are captured with a DSLR camera. Each scene consists of four images labeled as right view, left view, center view, and the all-in-focus ground truth. We train our network by taking the center view image as input and computing loss values between the predicted and ground-truth images. The training strategy follows that of previous algorithms (Cui, Tao, Bing, et al., 2023; Cui, Tao, Jing, & Knoll, 2023; Ruan, Chen, Li, & Lam, 2022).

4.2.3. Image desnowing

For this task, we use three datasets for evaluation, *i.e.,* CSD (Chen, Fang, et al., 2021), SRRS (Chen, Fang, Ding, Tsai, & Kuo, 2020), and Snow100K (Liu, Jaw, Huang, & Hwang, 2018). We train separate models for each dataset. The dataset setting is identical to previous algorithms (Cui, Ren, Cao, & Knoll, 2023; Cui, Tao, Jing, & Knoll, 2023) for a fair comparison, where we randomly select 2,500 image pairs from the training set for training and 2,000 image pairs from the test set for evaluation.

4.2.4. Image denoising

Following (Zamir et al., 2022), we train the network on a compound dataset, collected from DIV2K (Agustsson & Timofte, 2017), Flickr2K (Lim, Son, Kim, Nah, & Mu Lee, 2017), BSD500 (Arbelaez, Maire, Fowlkes, & Malik, 2010), and WED (Ma et al., 2016). Testing is performed on the BSD68 (Martin, Fowlkes, Tal, & Malik, 2001) dataset. The model is trained for 200 epochs with the initial learning rate as $1e^{-4}$. We train individual models for different noise levels ($\sigma \in \{15, 25, 50\}$).

4.3. Experimental results

4.3.1. Image dehazing

The results on daytime synthetic datasets are presented in Table 1. As can be seen, our method outperforms SANet (Cui, Tao, Jing, & Knoll, 2023) by 0.96 dB PSNR on SOTS-Indoor (Li et al., 2018) with additional 0.05M parameters and 0.46 GFLOPs, demonstrating the high efficiency of our frequency strip attention unit. Compared to the expensive Transformer-based algorithms, *i.e.,* DehazeFormer-L (Song et al., 2023) and DeHamer (Guo et al., 2022), our method produces significant performance gains with much fewer parameters and lower complexity. Additionally, our network outperforms the recent algorithm, FocalNet (Cui, Ren, Cao, & Knoll, 2023), by 0.54 dB and 0.68 dB PSNR on SOTS-Indoor and SOTS-Outdoor, respectively. The visual comparisons on the SOTS-Indoor dataset are illustrated in Fig. 4. We can conclude from the figure that our method is more effective in haze removal than other competitors.

Furthermore, we compare our method with state-of-the-art approaches on four widely used real-world datasets: NH-HAZE (Ancuti et al., 2020), NH-HAZE2 (Ancuti et al., 2021), O-Haze (Ancuti et al., 2018), and Dense-Haze (Ancuti et al., 2019). The results are reported

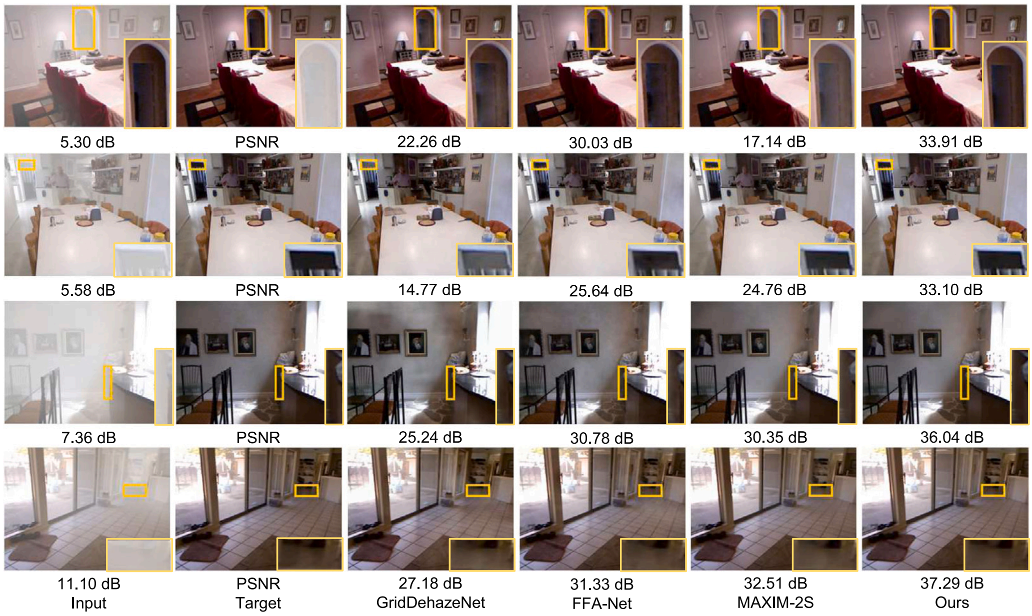

Fig. 4. Image dehazing comparisons on the SOTS-Indoor (Li et al., 2018) dataset.

Table 1
Image dehazing comparisons on SOTS-Indoor (Li et al., 2018) and SOTS-Outdoor (Li et al., 2018).

Method	SOTS-Indoor		SOTS-Outdoor		Params	FLOPs
	PSNR	SSIM	PSNR	SSIM	(M)	(G)
DCP (He, Sun, & Tang, 2010)	16.62	0.818	19.13	0.815	–	–
DehazeNet (Cai et al., 2016)	19.82	0.821	24.75	0.927	0.009	0.581
MSBDN (Dong et al., 2020)	33.67	0.985	33.48	0.982	31.35	41.54
FFA-Net (Qin et al., 2020)	36.39	0.989	33.57	0.984	4.456	287.8
AECR-Net (Wu et al., 2021)	37.17	0.990	–	–	2.611	52.20
MAXIM-2S (Tu et al., 2022)	38.11	0.991	34.19	0.985	14.1	108
PMNet (Ye et al., 2022)	38.41	0.990	34.74	0.985	18.90	81.13
DeHamer (Guo et al., 2022)	36.63	0.988	35.18	0.986	132.45	48.93
DehazeFormer-L (Song, He, Qian, & Du, 2023)	40.05	0.996	–	–	25.44	279.7
SANet (Cui, Tao, Jing, & Knoll, 2023)	40.40	0.996	38.01	**0.995**	3.81	37.26
FocalNet (Cui, Ren, Cao, & Knoll, 2023)	40.82	0.996	37.71	**0.995**	3.74	30.63
DSANet	**41.36**	**0.997**	**38.39**	**0.995**	3.86	37.72

Table 2
Image dehazing comparisons on four real-world datasets: NH-HAZE (Ancuti et al., 2020), NH-HAZE2 (Ancuti et al., 2021), O-Haze (Ancuti et al., 2018), and Dense-Haze (Ancuti et al., 2019).

Method	NH-HAZE		NH-HAZE2		O-Haze		Dense-Haze	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
DCP (He et al., 2010)	10.57	0.520	11.68	0.648	16.78	0.65	10.06	0.386
DehazeNet (Cai et al., 2016)	16.62	0.524	11.77	0.622	17.57	0.77	13.84	0.425
AOD-Net (Li, Peng, Wang, Xu, & Feng, 2017)	15.40	0.569	12.33	0.631	15.03	0.54	13.14	0.414
GDNet (Liu et al., 2019)	13.80	0.537	19.26	0.805	23.51	0.83	13.31	0.368
FFA-Net (Qin et al., 2020)	19.87	0.692	20.00	0.823	22.12	0.77	14.39	0.452
MSBDN (Dong et al., 2020)	19.23	0.706	20.11	0.800	24.36	0.75	15.37	0.486
AECR-Net (Wu et al., 2021)	19.88	0.717	20.68	0.828	–	–	15.80	0.466
SDCE (Zhu et al., 2023)	20.42	0.735	–	–	24.92	0.84	**16.85**	0.602
PMNet (Ye et al., 2022)	20.42	0.73	–	–	24.64	0.83	16.79	0.51
DSANet	**20.51**	**0.801**	**21.63**	**0.841**	**25.79**	**0.94**	16.70	**0.607**

Table 3
Nighttime image dehazing on the NHR (Zhang et al., 2020) dataset.

Method	GS Li, Tan, and Brown (2015)	MRPF Zhang, Cao, Fang, Kang, and Wen Chen (2017)	MRP Zhang et al. (2017)	OSFD Zhang et al. (2020)	HCD Wang, Tao, et al. (2022)	FocalNet Cui, Ren, Cao, and Knoll (2023)	DSANet Ours
PSNR	17.32	16.95	19.93	21.32	23.43	25.35	**28.08**
SSIM	0.629	0.667	0.777	0.804	0.953	0.969	**0.978**

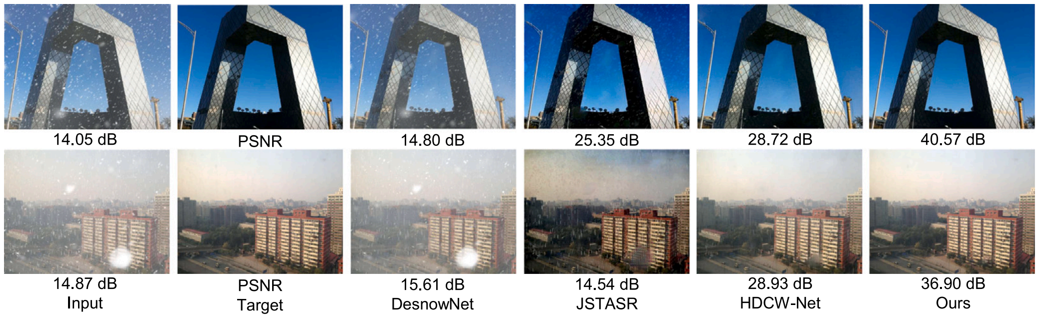

Fig. 5. Image desnowing comparisons on the CSD (Chen, Fang, et al., 2021) dataset.

Table 4
Image desnowing comparisons on three desnowing datasets: CSD (Chen, Fang, et al., 2021), SRRS (Chen et al., 2020), and Snow100K (Liu, Jaw, et al., 2018).

Method	CSD		SRRS		Snow100K	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
DesnowNet (Liu, Jaw, et al., 2018)	20.13	0.81	20.38	0.84	30.50	0.94
CycleGAN (Engin, Genc, & Kemal Ekenel, 2018)	20.98	0.80	20.21	0.74	26.81	0.89
All in One (Li, Tan, & Cheong, 2020)	26.31	0.87	24.98	0.88	26.07	0.88
JSTASR (Chen et al., 2020)	27.96	0.88	25.82	0.89	23.12	0.86
HDCW-Net (Chen, Fang, et al., 2021)	29.06	0.91	27.78	0.92	31.54	**0.95**
SMGARN (Cheng, Li, Chen, Zhang, & Zeng, 2022)	31.93	0.95	29.14	0.94	31.92	0.93
TransWeather (Valanarasu, Yasarla, & Patel, 2022)	31.76	0.93	28.29	0.92	31.82	0.93
SANet (Cui, Tao, Jing, & Knoll, 2023)	36.39	0.98	–	–	–	–
FocalNet (Cui, Ren, Cao, & Knoll, 2023)	37.18	**0.99**	31.34	**0.98**	33.53	**0.95**
IRNeXt (Cui, Ren, Yang, Cao, & Knoll, 2023)	37.29	**0.99**	31.91	**0.98**	33.61	**0.95**
DSANet	**38.09**	**0.99**	**31.97**	**0.98**	**33.70**	**0.95**

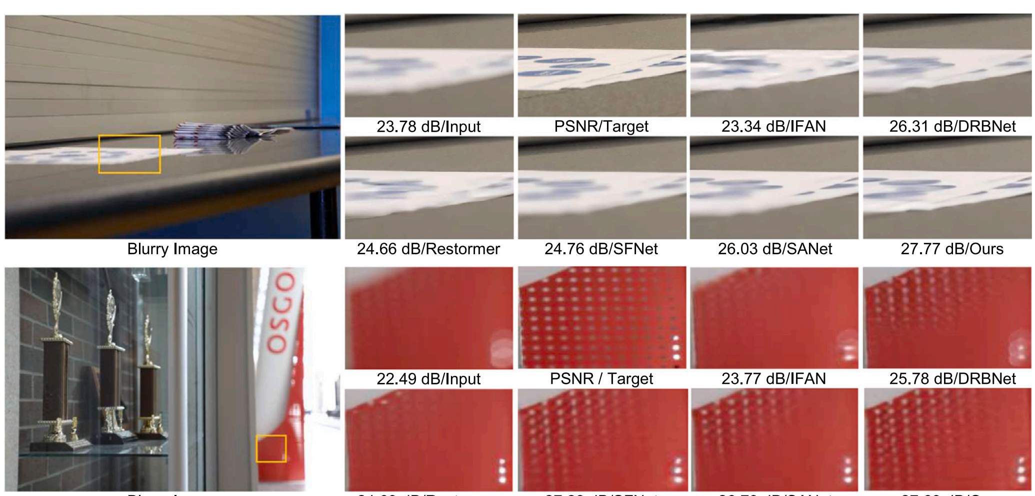

Fig. 6. Single-image defocus deblurring comparisons on the DPDD (Abuolaim & Brown, 2020) dataset.

in Table 2. As seen, our method obtains the best performance on most metrics. Specifically, our model is superior to PMNet (Ye et al., 2022) with performance gains of 0.09 dB and 1.15 dB PSNR on NH-HAZE and O-HAZE, respectively.

Apart from the daytime datasets, we also evaluate our model in nighttime scenes. Table 3 shows the numerical comparisons on the NHR (Zhang et al., 2020) dataset. Our method achieves a performance gain of 2.73 dB PSNR over the recent FocalNet (Cui, Ren, Cao, & Knoll, 2023) with only 0.12M more parameters and 7.09 G higher FLOPs.

4.3.2. *Image desnowing*

We verify the efficacy of our model on three desnowing datasets: CSD (Chen, Fang, et al., 2021), SRRS (Chen et al., 2020), and Snow100K (Liu, Jaw, et al., 2018). Table 4 shows that our framework performs well on all metrics. Specifically, compared to the recent FocalNet (Cui, Ren, Cao, & Knoll, 2023), our method achieves accuracy gains of 0.63 dB and 0.17 dB in terms of PSNR on the SRRS (Chen et al., 2020) and Snow100K (Liu, Jaw, et al., 2018) datasets, respectively. Furthermore, on the more complicated CSD (Chen, Fang, et al., 2021)

Fig. 7. Grayscale image denoising comparisons on the BSD68 (Martin et al., 2001) dataset.

Table 5
Single-image defocus deblurring comparisons on the DPDD (Abuolaim & Brown, 2020) dataset. FLOPs are measured on a $3 \times 720 \times 1280$ patch.

Method	Indoor scenes				Outdoor scenes				Combined				Params	FLOPs
	PSNR↑	SSIM↑	MAE↓	LPIPS↓	PSNR	SSIM	MAE	LPIPS	PSNR	SSIM	MAE	LPIPS	(M)	(G)
DPDNet (Abuolaim & Brown, 2020)	26.54	0.816	0.031	0.239	22.25	0.682	0.056	0.313	24.34	0.747	0.044	0.277	31.03	770
MDP (Abuolaim, Afifi, & Brown, 2022)	28.02	0.841	0.027	–	22.82	0.690	0.052	–	25.35	0.763	0.040	–	46.86	1898
DRBNet (Ruan et al., 2022)			–				–		25.73	0.791	–	0.183	11.69	693
SFNet (Cui, Tao, Bing, et al., 2023)	29.16	0.878	**0.023**	0.168	23.45	0.747	**0.049**	0.244	26.23	0.811	0.037	0.207	13.24	1763.84
LaKDNet (Ruan, Bemana, Seidel, Myszkowski, & Chen, 2023)			–				–		26.15	0.810	–	**0.155**	17.7	1208
Restormer (Zamir et al., 2022)	28.87	**0.882**	0.025	**0.145**	23.24	0.743	0.050	**0.209**	25.98	0.811	0.038	0.178	26.13	1983
FocalNet (Cui, Ren, Cao, & Knoll, 2023)	**29.10**	0.876	0.024	0.173	23.41	0.743	**0.049**	0.246	26.18	0.808	0.037	0.210	12.82	1376.43
SANet (Cui, Tao, Jing, & Knoll, 2023)	**29.30**	0.878	0.024	0.163	23.43	0.748	**0.049**	0.227	26.29	0.811	0.037	0.196	13.11	1750.74
DSANet	29.27	0.881	0.024	0.158	**23.50**	**0.749**	**0.049**	0.231	**26.31**	**0.813**	**0.036**	0.195	13.16	1757.13

Table 6
Gaussian grayscale image denoising results on the BSD68 (Martin et al., 2001) dataset. σ denotes the noise level.

Method	NLRN Liu, Wen, Fan, Loy, and Huang (2018)	DeamNet Ren, He, Wang, and Zhao (2021)	DAGL Mou, Zhang, and Wu (2021)	SwinIR Liang et al. (2021)	Restormer Zamir et al. (2022)	DSANet Ours
σ=15	31.88	31.91	31.93	**31.97**	31.96	**31.97**
σ=25	29.41	29.44	29.46	29.50	29.52	**29.54**
σ=50	26.47	26.54	26.51	26.58	26.62	**26.65**
FLOPs (G)	–	146	256	759	141	139

dataset, the advantage becomes more obvious with a performance gain of 0.91 dB PSNR. Additionally, our method outperforms IRNeXt (Cui, Ren, Yang, et al., 2023) on all three datasets in terms of PSNR. We provide visual comparisons in Fig. 5. We can see that our results are much visually closer to the ground truth images. The results suggest the effectiveness of our method.

4.3.3. Image defocus deblurring

We conduct experiments for single-image defocus deblurring on the commonly adopted DPDD (Abuolaim & Brown, 2020) dataset. Table 5 shows that our method performs better than the single-domain method SANet (Cui, Tao, Jing, & Knoll, 2023) on most metrics with comparable computation overhead. More specifically, in the outdoor scenes, our model yields a performance gain of 0.07 dB PSNR over SANet (Cui, Tao, Jing, & Knoll, 2023). Moreover, the proposed network also significantly outperforms the strong Transformer-based method Restormer (Zamir et al., 2022) by 0.33 dB in the combined category with 11% lower complexity and half parameters. Additionally, compared to FocalNet (Cui, Ren, Cao, & Knoll, 2023), our DSANet achieves a remarkable gain of 0.13 dB PSNR in the combined category with comparable parameters. The visual results are illustrated in Fig. 6. Compared with other state-of-the-art algorithms, our method recovers more detailed information from blurry observations and generates more visually-faithful images.

4.3.4. Image denoising

The grayscale image denoising results are presented in Table 6. Our method outperforms the strong Transformer-based Restormer (Zamir et al., 2022) for all noise levels on the BSD68 (Martin et al., 2001) dataset. Furthermore, our DSANet is superior to SwinIR (Liang et al., 2021) by achieving a significant performance gain of 0.07 dB PSNR for $\sigma = 50$. The visual comparisons between our method and Restormer (Zamir et al., 2022) are presented in Fig. 7. The images produced by our model are much sharper.

4.3.5. Evaluation on real data

To demonstrate the robustness of our network in real scenarios, we directly apply the pre-trained model on RESIDE-Outdoor (Li et al., 2018) to real-world images obtained from UAVDT (Du et al., 2018). The dehazed images are presented in Fig. 8. The results suggest that our method exhibits a potential to deal with real-world degraded images, though trained on synthetic data.

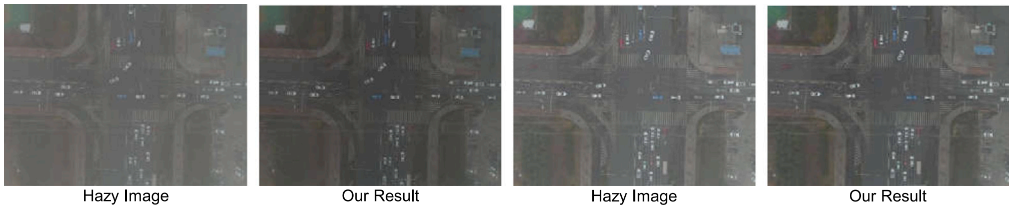

Fig. 8. Evaluation on real-world hazy images of UAVDT (Du et al., 2018). Our model is pre-trained on RESIDE-Outdoor (Li et al., 2018).

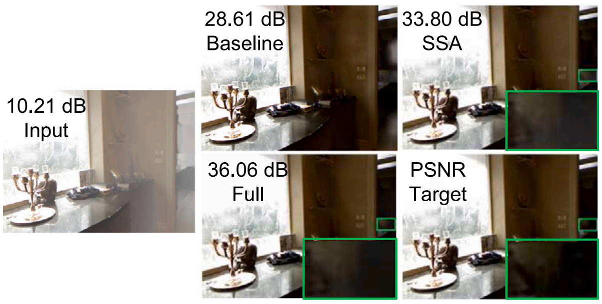

Fig. 9. Visual comparisons for ablation studies.

Table 7
Ablation studies towards better performance for the proposed components. SSA-V=Spatial vertical strip attention. SSA-V-H denotes using spatial vertical and horizontal strip attention successively. *Parallel* combines the output of two-directional strip attention operators via addition.

No.	Method	PSNR	SSIM	FLOPs/G
1	Baseline	31.33	0.9836	15.44
2	SSA-V	34.63	0.9896	15.52
3	SSA-H	34.67	0.9897	15.52
4	SSA-Parallel	35.28	0.9909	15.58
5	SSA-V-H	35.82	0.9914	15.58
6	SSA-H-V	35.89	0.9918	15.58
7	FSA-V+SSA	36.63	0.9924	15.88
8	FSA-H+SSA	36.81	0.9928	15.88
9	FSA-Parallel+SSA	37.14	0.9928	15.91
10	FSA-V-H+SSA	37.17	0.9928	15.91
11	FSA-H-V+SSA	37.18	0.9929	15.91

4.4. Ablation studies

In this section, we conduct ablation studies to demonstrate the effectiveness of the proposed components and investigate different design choices. All models are trained on the RESIDE-Indoor (Li et al., 2018) dataset for 300 epochs with $N = 0$ and tested on SOTS-Indoor (Li et al., 2018). Other configurations remain identical to that of the final dehazing model. The baseline model is obtained by removing the DSAM from our model.

4.4.1. Improvements of the proposed components

Table 7 shows that the baseline model achieves 31.33 dB PSNR on the SOTS-Indoor dataset. The proposed spatial vertical and horizontal strip attention operators obtain performance gains of 3.3 dB and 3.34 dB PSNR, respectively, over the baseline model with negligible introduced complexity. Then, we study the combination pattern of the two-directional spatial strip attention. All three versions (Table 7 (4-6)) produce higher scores than the models using single-directional attention, demonstrating the compatibility of vertical and horizontal attention. Among these three variants, the model deploying horizontal and vertical attention successively yields the best result, 4.56 dB PSNR higher than the baseline model. Therefore, we choose this setting in our final model for better performance.

Table 8
Design choices for FSA.

Method	Average pooling	Max pooling
PSNR	37.18	37.02

Table 9
Design choices for DSAM.

Method	FSA+SSA	SSA+FSA	SSA+FSA-Parallel	FSA+FSA	SSA+SSA
PSNR	37.18	37.06	36.84	37.08	36.41

Table 10
Ablation studies for different strip sizes. FSA denotes the default setting in our final model where horizontal and vertical frequency strip operators are used in order. K7 means $K = 7$. *Global* means applying the global strip-shaped receptive field.

Method	PSNR	SSIM	FLOPs/G
SSA	35.89	0.9918	15.58
SSA+FSA-K7	36.49	0.9924	15.88
SSA+FSA-K11	36.67	0.9925	15.88
SSA+FSA-K7-K11	36.84	0.9927	15.89
SSA+FSA-K7-K11-Global	37.18	0.9929	15.91

We further study the efficacy of FSA. The vertical and horizontal versions lead to 0.74 dB and 0.92 dB PSNR performance gains over the model with SSA (Table 7 (6)). Using two-directional spectral strip attention operators in parallel produces 37.14 dB PSNR, which is higher than using a single one. We finally select the version of Table 7 (11) as the default setting in our model.

The visual comparisons for ablation studies are presented in Fig. 9. The model with SSA is more effective than the baseline model in removing haze degradations. Furthermore, the result generated by the full model (Table 7 (11)) is much closer to the ground truth.

4.4.2. Design choices for FSA

We utilize average pooling techniques in FSA to obtain the lowest frequency components. In this part, we conduct experiments by substituting average pooling with max pooling. Table 8 shows that our model outperforms the alternative by 0.16 dB PSNR, demonstrating the effectiveness of our design.

4.4.3. Design choices for DSAM

We investigate different design choices for DSAM in Table 9. We first study the impact of the deployment order of FSA and SSA. As seen, utilizing FSA and SSA in series leads to 37.18 dB PSNR on the SOTS-Indoor (Li et al., 2018) dataset. Changing the deployment order decreases the performance by 0.12 dB PSNR. The parallel combination version achieves the worst result. We then verify the indispensability of each module by using a single one twice. Employing two FSA or SSA yields inferior performance than ours. These results suggest the efficacy of our design.

4.4.4. Different receptive fields

We further study the influence of using different strip sizes by taking FSA as an example. Table 10 shows that FSA with $K = 7$ obtains a performance gain of 0.6 dB PSNR over the model only using SSA. Enlarging

Fig. 10. Our results on the NH-HAZE (Ancuti et al., 2020) dataset.

Table 11
Design choices for loss functions.

Method	Frequency loss	Spatial loss	Dual-domain loss
PSNR	29.00	30.71	31.33

the receptive field by setting $K = 11$ produces a further improvement. Simultaneously using two different receptive fields generates a higher score than models using a single receptive field. The complete model with an additional global strip-shaped receptive field performs best. The results suggest the effectiveness of our multi-scale design.

4.4.5. Design choices for loss functions

In this part, we conduct experiments to verify the efficacy of dual-domain loss functions. The results are shown in Table 11. We can see that using only the frequency loss function achieves 29.00 dB PSNR, while the spatial loss improves the performance to 30.71 dB, which suggests the more important role of spatial loss in image restoration. Utilizing dual-domain loss functions in our model produces 31.33 dB PSNR, which is higher than both two single-domain versions.

5. Limitation

Despite more advanced performance on the real-world NH-HAZE (Ancuti et al., 2020) dataset, our method still struggles to fully remove hazy degradations due to the limited training data. As illustrated in Fig. 10, there are remarkable gaps between our results and reference images. A promising direction is to develop domain adaptation algorithms to excavate the potential utility of synthetic datasets for real-world scenarios.

6. Conclusion

In this paper, we present a dual-domain strip attention network, termed DSANet, for efficient image restoration. Specifically, the SSA generates attention weights and performs two-directional information aggregation within strip regions in a cheap convolution manner. Moreover, the FSA refines spectra by frequency decomposition and modulation using extremely simple strip pooling techniques. To enhance multi-scale representation learning, the DSAM inserts different strip sizes in both SSA and FSA. Finally, the proposed network achieves state-of-the-art performance on 12 different datasets for four representative image restoration tasks, including image dehazing, image desnowing, image denoising, and image defocus deblurring.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Abuolaim, A., Afifi, M., & Brown, M. S. (2022). Improving single-image defocus deblurring: How dual-pixel images help through multi-task learning. In *Proceedings of the IEEE winter conference on applications of computer vision* (pp. 1231–1239).

Abuolaim, A., & Brown, M. S. (2020). Defocus deblurring using dual-pixel data. In *Proceedings of the European conference on computer vision* (pp. 111–126).

Agustsson, E., & Timofte, R. (2017). Ntire 2017 challenge on single image super-resolution: Dataset and study. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops* (pp. 126–135).

Ancuti, C. O., Ancuti, C., Sbert, M., & Timofte, R. (2019). Dense-haze: A benchmark for image dehazing with dense-haze and haze-free images. In *IEEE international conference on image processing* (pp. 1014–1018).

Ancuti, C. O., Ancuti, C., & Timofte, R. (2020). NH-HAZE: An image dehazing benchmark with non-homogeneous hazy and haze-free images. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*.

Ancuti, C. O., Ancuti, C., Timofte, R., & De Vleeschouwer, C. (2018). O-haze: a dehazing benchmark with real hazy and haze-free outdoor images. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops* (pp. 754–762).

Ancuti, C. O., Ancuti, C., Vasluianu, F.-A., & Timofte, R. (2021). NTIRE 2021 nonhomogeneous dehazing challenge report. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 627–646).

Arbelaez, P., Maire, M., Fowlkes, C., & Malik, J. (2010). Contour detection and hierarchical image segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *33*(5), 898–916.

Cai, B., Xu, X., Jia, K., Qing, C., & Tao, D. (2016). Dehazenet: An end-to-end system for single image haze removal. *IEEE Transactions on Image Processing*, *25*(11), 5187–5198.

Chen, L., Chu, X., Zhang, X., & Sun, J. (2022). Simple baselines for image restoration. In *Proceedings of the European conference on computer vision*.

Chen, Y., Fan, H., Xu, B., Yan, Z., Kalantidis, Y., Rohrbach, M., et al. (2019). Drop an octave: Reducing spatial redundancy in convolutional neural networks with octave convolution. In *Proceedings of the IEEE international conference on computer vision* (pp. 3435–3444).

Chen, W.-T., Fang, H.-Y., Ding, J.-J., Tsai, C.-C., & Kuo, S.-Y. (2020). JSTASR: Joint size and transparency-aware snow removal algorithm based on modified partial convolution and veiling effect removal. In *Proceedings of the European conference on computer vision* (pp. 754–770).

Chen, W.-T., Fang, H.-Y., Hsieh, C.-L., Tsai, C.-C., Chen, I., Ding, J.-J., et al. (2021). All snow removed: Single image desnowing algorithm using hierarchical dual-tree complex wavelet representation and contradict channel loss. In *Proceedings of the IEEE international conference on computer vision* (pp. 4196–4205).

Chen, H., Wang, Y., Guo, T., Xu, C., Deng, Y., Liu, Z., et al. (2021). Pre-trained image processing transformer. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 12299–12310).

Chen, Z., Zhang, Y., Gu, J., Kong, L., Yuan, X., et al. (2022). Cross aggregation transformer for image restoration. *Advances in Neural Information Processing Systems*, *35*, 25478–25490.

Cheng, B., Li, J., Chen, Y., Zhang, S., & Zeng, T. (2022). Snow mask guided adaptive residual network for image snow removal. arXiv preprint arXiv:2207.04754.

Cho, S.-J., Ji, S.-W., Hong, J.-P., Jung, S.-W., & Ko, S.-J. (2021). Rethinking coarse-to-fine approach in single image deblurring. In *Proceedings of the IEEE international conference on computer vision* (pp. 4641–4650).

Cui, Y., & Knoll, A. (2023). PSNet: Towards efficient image restoration with self-attention. *IEEE Robotics and Automation Letters*.

Cui, Y., Ren, W., Cao, X., & Knoll, A. (2023). Focal network for image restoration. In *Proceedings of the IEEE international conference on computer vision*.

Cui, Y., Ren, W., Yang, S., Cao, X., & Knoll, A. (2023). IRNeXt: Rethinking convolutional network design for image restoration. In *International conference on machine learning*.

Cui, Y., Tao, Y., Bing, Z., Ren, W., Gao, X., Cao, X., et al. (2023). Selective frequency network for image restoration. In *International conference on learning representations*.

Cui, Y., Tao, Y., Jing, L., & Knoll, A. (2023). Strip attention for image restoration. In *Proceedings of the international joint conferences on artificial intelligence*.

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). Imagenet: A large-scale hierarchical image database. In *Proceedings IEEE conference on computer vision and pattern recognition* (pp. 248–255).

Dong, X., Bao, J., Chen, D., Zhang, W., Yu, N., Yuan, L., et al. (2022). Cswin transformer: A general vision transformer backbone with cross-shaped windows. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 12124–12134).

Dong, H., Pan, J., Xiang, L., Hu, Z., Zhang, X., Wang, F., et al. (2020). Multi-scale boosted dehazing network with dense feature fusion. In *Proceedings of the IEEE conference on computer vision and pattern recognition*.

Du, D., Qi, Y., Yu, H., Yang, Y., Duan, K., Li, G., et al. (2018). The unmanned aerial vehicle benchmark: Object detection and tracking. In *Proceedings of the European conference on computer vision* (pp. 370–386).

Engin, D., Genc, A., & Kemal Ekenel, H. (2018). Cycle-dehaze: Enhanced cyclegan for single image dehazing. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*.

Fu, X., Huang, J., Ding, X., Liao, Y., & Paisley, J. (2017). Clearing the skies: A deep network architecture for single-image rain removal. *IEEE Transactions on Image Processing*, *26*(6), 2944–2956.

Guo, C.-L., Yan, Q., Anwar, S., Cong, R., Ren, W., & Li, C. (2022). Image dehazing transformer with transmission-aware 3D position embedding. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 5812–5820).

Guo, S., Yong, H., Zhang, X., Ma, J., & Zhang, L. (2023). Spatial-frequency attention for image denoising. arXiv preprint arXiv:2302.13598.

He, K., Sun, J., & Tang, X. (2010). Single image haze removal using dark channel prior. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *33*(12), 2341–2353.

Huang, Y., Huang, J., Liu, J., Dong, Y., Lv, J., & Chen, S. (2023). Wavedm: Wavelet-based diffusion models for image restoration. arXiv preprint arXiv:2305.13819.

Huang, J., Liu, Y., Zhao, F., Yan, K., Zhang, J., Huang, Y., et al. (2022). Deep Fourier-based exposure correction network with spatial-frequency interaction. In *Proceedings of the European conference on computer vision* (pp. 163–180).

Lee, J., Son, H., Rim, J., Cho, S., & Lee, S. (2021). Iterative filter adaptive network for single image defocus deblurring. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2034–2042).

Li, Y., Fan, Y., Xiang, X., Demandolx, D., Ranjan, R., Timofte, R., et al. (2023). Efficient and explicit modelling of image hierarchies for image restoration. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 18278–18289).

Li, C., Guo, C., Zhou, M., Liang, Z., Zhou, S., Feng, R., et al. (2023). Embedding Fourier for ultra-high-definition low-light image enhancement. In *The eleventh international conference on learning representations*. URL https://openreview.net/forum?id=5N0wtJZ89r9.

Li, B., Peng, X., Wang, Z., Xu, J., & Feng, D. (2017). AOD-net: All-in-one dehazing network. In *Proceedings of the IEEE international conference on computer vision*.

Li, B., Ren, W., Fu, D., Tao, D., Feng, D., Zeng, W., et al. (2018). Benchmarking single-image dehazing and beyond. *IEEE Transactions on Image Processing*, *28*(1), 492–505.

Li, Y., Tan, R. T., & Brown, M. S. (2015). Nighttime haze removal with glow and multiple light colors. In *Proceedings of the IEEE international conference on computer vision* (pp. 226–234).

Li, R., Tan, R. T., & Cheong, L.-F. (2020). All in one bad weather removal using architectural search. In *Proceedings of the IEEE conference on computer vision and pattern recognition*.

Liang, J., Cao, J., Sun, G., Zhang, K., Van Gool, L., & Timofte, R. (2021). Swinir: Image restoration using swin transformer. In *Proceedings of the IEEE international conference on computer vision* (pp. 1833–1844).

Lim, B., Son, S., Kim, H., Nah, S., & Mu Lee, K. (2017). Enhanced deep residual networks for single image super-resolution. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops* (pp. 136–144).

Liu, Y.-F., Jaw, D.-W., Huang, S.-C., & Hwang, J.-N. (2018). DesnowNet: Context-aware deep network for snow removal. *IEEE Transactions on Image Processing*, *27*(6), 3064–3073.

Liu, X., Ma, Y., Shi, Z., & Chen, J. (2019). Griddehazenet: Attention-based multi-scale network for image dehazing. In *Proceedings of the IEEE international conference on computer vision* (pp. 7314–7323).

Liu, D., Wen, B., Fan, Y., Loy, C. C., & Huang, T. S. (2018). Non-local recurrent network for image restoration. *Advances in Neural Information Processing Systems*, *31*.

Liu, K.-H., Yeh, C.-H., Chung, J.-W., & Chang, C.-Y. (2020). A motion deblur method based on multi-scale high frequency residual image learning. *IEEE Access*, *8*, 66025–66036.

Ma, K., Duanmu, Z., Wu, Q., Wang, Z., Yong, H., Li, H., et al. (2016). Waterloo exploration database: New challenges for image quality assessment models. *IEEE Transactions on Image Processing*, *26*(2), 1004–1016.

Mao, X., Liu, Y., Shen, W., Li, Q., & Wang, Y. (2021). Deep residual fourier transformation for single image deblurring. arXiv preprint arXiv:2111.11745.

Martin, D., Fowlkes, C., Tal, D., & Malik, J. (2001). A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. In *Proceedings eighth IEEE international conference on computer vision* (pp. 416–423).

Mou, C., Zhang, J., & Wu, Z. (2021). Dynamic attentive graph learning for image restoration. In *Proceedings of the IEEE international conference on computer vision* (pp. 4328–4337).

Nah, S., Hyun Kim, T., & Mu Lee, K. (2017). Deep multi-scale convolutional neural network for dynamic scene deblurring. In *Proceedings of the IEEE conference on computer vision and pattern recognition*.

Narasimhan, S. G., & Nayar, S. K. (2000). Chromatic framework for vision in bad weather. In *Proceedings IEEE conference on computer vision and pattern recognition, vol. 1* (pp. 598–605).

Qin, X., Wang, Z., Bai, Y., Xie, X., & Jia, H. (2020). FFA-Net: Feature fusion attention network for single image dehazing. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 11908–11915).

Ren, C., He, X., Wang, C., & Zhao, Z. (2021). Adaptive consistency prior based deep network for image denoising. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 8596–8606).

Ruan, L., Bemana, M., Seidel, H.-p., Myszkowski, K., & Chen, B. (2023). Revisiting image deblurring with an efficient ConvNet. arXiv preprint arXiv:2302.02234.

Ruan, L., Chen, B., Li, J., & Lam, M. (2022). Learning to deblur using light field generated and real defocus images. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 16304–16313).

Song, Y., He, Z., Qian, H., & Du, X. (2023). Vision transformers for single image dehazing. *IEEE Transactions on Image Processing*, *32*, 1927–1941.

Tian, C., Fei, L., Zheng, W., Xu, Y., Zuo, W., & Lin, C.-W. (2020). Deep learning on image denoising: An overview. *Neural Networks*, *131*, 251–275.

Tian, C., Xu, Y., Li, Z., Zuo, W., Fei, L., & Liu, H. (2020). Attention-guided CNN for image denoising. *Neural Networks*, *124*, 117–129.

Tian, C., Xu, Y., & Zuo, W. (2020). Image denoising using deep CNN with batch renormalization. *Neural Networks*, *121*, 461–473.

Tsai, F.-J., Peng, Y.-T., Lin, Y.-Y., Tsai, C.-C., & Lin, C.-W. (2022). Stripformer: Strip transformer for fast image deblurring. In *Proceedings of the European conference on computer vision*.

Tu, Z., Talebi, H., Zhang, H., Yang, F., Milanfar, P., Bovik, A., et al. (2022). MAXIM: Multi-axis MLP for image processing. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 5769–5780).

Valanarasu, J. M. J., Yasarla, R., & Patel, V. M. (2022). TransWeather: Transformer-based restoration of images degraded by adverse weather conditions. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2353–2363).

Wang, Z., Cun, X., Bao, J., Zhou, W., Liu, J., & Li, H. (2022). Uformer: A general U-shaped transformer for image restoration. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 17683–17693).

Wang, T., Tao, G., Lu, W., Zhang, K., Luo, W., Zhang, X., et al. (2022). Restoring vision in hazy weather with hierarchical contrastive learning. arXiv preprint arXiv:2212.11473.

Wu, H., Qu, Y., Lin, S., Zhou, J., Qiao, R., Zhang, Z., et al. (2021). Contrastive learning for compact single image dehazing. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 10551–10560).

Ye, T., Zhang, Y., Jiang, M., Chen, L., Liu, Y., Chen, S., et al. (2022). Perceiving and modeling density for image dehazing. In *Proceedings of the European conference on computer vision* (pp. 130–145).

Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., & Yang, M.-H. (2022). Restormer: Efficient transformer for high-resolution image restoration. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 5728–5739).

Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., Yang, M.-H., et al. (2021). Multi-stage progressive image restoration. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 14821–14831).

Zhang, J., Cao, Y., Fang, S., Kang, Y., & Wen Chen, C. (2017). Fast haze removal for nighttime image using maximum reflectance prior. In *Proceedings of the IEEE conference on computer vision and pattern recognition*.

Zhang, J., Cao, Y., Zha, Z.-J., & Tao, D. (2020). Nighttime dehazing with a synthetic benchmark. In *Proceedings of the ACM international conference on multimedia* (pp. 2355–2363).

Zhang, Y., Li, Q., Qi, M., Liu, D., Kong, J., & Wang, J. (2022). Multi-scale frequency separation network for image deblurring. arXiv preprint arXiv:2206.00798.

Zhang, K., Ren, W., Luo, W., Lai, W.-S., Stenger, B., Yang, M.-H., et al. (2022). Deep image deblurring: A survey. *International Journal of Computer Vision*, *130*(9), 2103–2130.

Zhang, X., Wang, J., Wang, T., & Jiang, R. (2021). Hierarchical feature fusion with mixed convolution attention for single image dehazing. *IEEE Transactions on Circuits and Systems for Video Technology*, *32*(2), 510–522.

Zhu, Z., Zhang, D., Wang, Z., Feng, S., & Duan, P. (2023). Spectral dual-channel encoding for image dehazing. *IEEE Transactions on Circuits and Systems for Video Technology*.

Zou, W., Jiang, M., Zhang, Y., Chen, L., Lu, Z., & Wu, Y. (2021). Sdwnet: A straight dilated network with wavelet transformation for image deblurring. In *Proceedings of the IEEE international conference on computer vision workshops* (pp. 1895–1904).